A Manager/Worker POMDP Decomposition of the G1 Meta-Reasoning Stack

Adrián Lendínez Ibáñez — working note in response to RQ, 21 July 2026. Goal: the list of high-level and low-level states, observations and actions we agreed to draft, with each element mapped to its already-implemented counterpart in the G1 stack and to the logged fields that would ground the particle-filter probabilities. Everything in code font is a field recorded per run today.

1. The framing in one paragraph

Low-level policies are the **analogies** — each a capability-scoped skill (multi-parameter profile {v_cap, turn_cap, robot_r} + engagement subgoals) documented in the Capability Map. The **Manager POMDP** is the meta-reasoner: it holds beliefs over regime variables it cannot observe directly (payload state, environment regime, sensing health, analogy validity, progress), and its actions set the context the Worker runs in — which analogy, which intermediate subgoal, which parameter bounds, which sensor to trust, when to explore, when to stop and ask. The **Worker POMDP** is the per-analogy executor: beliefs over the local obstacle field and own pose, actions = trajectories and recovery primitives. The Worker runs as an option (semi-MDP) until the Manager changes context — RQ's Manager/Worker split is exactly this options architecture.

2. Manager POMDP (meta-reasoner, 0.5–2 Hz)

2a. Hidden state variables and their observable proxies

Hidden state (belief over...)	Observable proxies (logged)	Current belief machinery
Payload state: lid open/closed, fill level, settled vs fresh cup	spill marks (`spills_human`, UDP GT), spill rate, time since last spill	Fragility channel: exponential dips + repeat load (platform-calibrated, `G1_M2_REALCAL`)
Environment regime: open corridor / door zone / tight pocket	`c0`, `c0_hard`, `clearL_m/clearR_m`, median clearance, door distance	Layer 3 relevance EMA (`meta2_env`, clamp 0.85–1.15)
Sensing health: laser trustworthy? camera needed? localization ok?	`laser_noise`, `filt_rej`, `scan_churn`, `loc_conf`, `loc_match`, `reloc_rate10s`, `color_near`, `perc_age`	SensingMonitor reliability -> per-reading DST uncertainty (`meta2_unc`)
Platform resources: battery sufficient? thermal/mechanical health?	`bat`, `cpuT`, per-joint telemetry (1 Hz track), IMU	Battery reading into reasoner; no belief dynamics yet (gap)
Analogy validity: is the active low-level policy still matched?	fulfilment `meta2_ful`, tension `meta2_tens`, spills under this analogy, QoE region occupancy	Layer 2 DST per analogy {m_match, m_mismatch, Θ}; Pl thresholds 0.45/0.70
Progress state: advancing / stuck / looping (a STATE, as Adrián noted — the action it licenses is exploration or escalation)	progress rate, path/goal distance history, collision events, P9 windows	Escalation windows (`G1_M2_ABORT_*`); aggressive-mode trigger
Decision robustness: how safe is the current arbitration itself?	top-2 fulfilment margin, uncertainty gap, attention mismatch	ρ_DCA per decision (`meta2_rho`)

2b. High-level ACTIONS (the Manager list)

#	Manager action	Implemented as	Status
M1	Select / switch low-level policy (analogy)	KEEP/SWITCH/FALLBACK over Cautious/Efficient with DST deploy veto	DONE (twin + real)
M2	Set intermediate subgoal for the analogy	Door-approach variant (Direct 0.85 m / Far 1.20 m / BiasPlus +0.12 m) parametrises the engagement FSM	DONE at the door; generalisation = next research line (B-pocket

			entry pose)
M3	Bound the Worker (parameter governance)	Profile caps {v, turn, robot_r}; L3 modulates with environment relevance; graded spill->speed coupling	DONE
M4	Arbitrate sensors (lidar vs camera)	COLOR-BRAKE: camera moderates when laser is height-band blind; yaw-gate: distrust laser while turning; DOORSTICKY: trust static map over flickering returns near the door	DONE as reflex rules; formalising as belief-conditioned action is the PF opportunity
M5	Explore vs exploit	G1_M2_DOOR_EXPLORE: least-tried variant forced in the TWIN (failure is free); hardware always exploits by Pl	DONE today — results in §4
M6	Reuse / seed skills	Ext C sim-to-real trust transfer (G1_M2_STATE_INIT, task-id remap); L4 start-up re-selection	DONE
M7	Escalate to human (HELP) / governed stop	P9 escalation; HELP sustained -> stop (the 23 s governed stop vs 513 s ungoverned loop pair)	DONE
M8	Refuse / renegotiate task on resources	Battery observed but no return-to-base / refusal policy	GAP (candidate M-action)

3. Worker POMDP (per-analogy executor, 3–10 Hz)

3a. Hidden state and observations

Hidden state (belief over...)	Observable proxies (logged)	Current belief machinery
Local obstacle field — including what the laser cannot see (below 0.7 m band; thin jambs edge-on)	laser cells per scan (`laser_snapshots` now with pose+motion phase), camera clamp columns, `color_rmin`, IMU contact	K-of-N persistence filter + per-cell score (`oscore`) = a discrete occupancy filter; DOORSTICKY freezes known-static cells
Own pose quality	SLAM pose, `loc_conf`, `loc_match`, reloc jumps	Gate on start (GATE_M); jump detection; no full pose posterior (PF opportunity)
Trajectory feasibility ("is a feasible trajectory?")	DWA candidate scoring vs clearance, `plan_n` (A* success), carrot errors	DWA + A* replan (the MDP solver RQ names)
Contact risk of own envelope	`c0` vs governed `robot_r`, `clearL/R` balance	Governed body radius (was the mechanism-A failure when fixed at 0.22)

3b. Low-level ACTIONS (the Worker list)

#	Worker action	Implemented as
W1	Trajectory selection within bounds	DWA velocity commands under profile caps
W2	Global replan	A* on static+confirmed costmap
W3	Engagement primitives	Door FSM: align -> strafe -> cross (subgoal set by M2)
W4	Recovery primitives	Back-off, rotate, ESCAPE (never governed — safety floor)
W5	Reflex guards	HARD-GUARD stop/slow on persistent walls; COLOR-BRAKE creep/stop (camera-only); VCAP
W6	Epistemic actions on its own map	Insert/decay cells, yaw-gate freeze while turning, DOORSTICKY fixation — actions taken to manage belief, not motion

4. Exploration as a Manager action: first result (digital twin)

12-run twin arm with forced least-tried rotation: 4 trials per variant (vs 8/0/0 under pure exploitation in the formal A/B). All three variants now carry grounded trust (Θ shrank from 0.50 to 0.12–0.16). Door_Direct confirmed as incumbent with full evidence (4/4 crossings, 0 engagement fails); Door_Far and Door_BiasPlus each took one engagement fail; BiasPlus additionally suffered two post-door pocket aborts — correctly charged to the pocket, not to the door phase (credit assignment respects the control envelope, the July-14 lesson). The ranking cost three aborted twin runs — free — and ships as meta2_state_twin_explored.json, the recommended G1_M2_STATE_INIT seed for real campaign 2.

Variant	Trials	Crossings	Fails	Final Pl	Mean reached time (s)
Door_Direct	4	4	0	1.00	93.6
Door_Far	4	3	1	0.93	103.0
Door_BiasPlus	4	3	1	0.93	107.4 (1/4 runs reached: 2 post-door pocket aborts)

The populated library seeds the real robot through Ext C (`G1_M2_STATE_INIT`): **exploration happens where failure is free; the hardware only ever exploits.** The resulting state file ships with the repo for campaign 2.

5. Where the particle filter fits — and what already grounds it

- **Correspondence.** The DST masses are already a belief: Bel = m_match is the pessimistic posterior, Pl = m_match + Θ the optimistic one, and Dempster's rule the evidence update. A particle filter adds what DST lacks: TEMPORAL dynamics (transition models for regime variables — cup settling, battery drain, sensing degradation while turning) and joint posteriors over correlated hidden states.
- **Worker side.** The K-of-N persistence filter + cell scores are a per-cell binary Bayes filter; a PF over (pose, local map) would unify it with localization quality — RQ's "Particle Filters and MDP solver" is literally this plus the existing DWA/A*.
- **Calibration corpus already logged.** Every observable named in §2–3 is recorded per run: 140+ twin runs and 60+ real runs, including per-sample DST uncertainties (`meta2_unc`), laser snapshots with pose+motion phase (for the emission model, noise +65% while turning), spill ground truth, and per-run outcomes. Defining the transition/emission models and fitting them to this corpus is a data-analysis task, not a new experimental campaign.

Proposed next step: we draft the formal tuple ⟨S, A, O, T, R⟩ for the Manager with 5–6 discrete regime variables (payload, environment, sensing, resource, validity, progress), reward = the existing meta-attention weights (progression/safety/fragility/battery/mobility — they are already the reward vector in the configs), and fit T/O from the logged campaigns; the Worker keeps its current solvers with the PF formalisation of its filters. That yields RQ's "complete framework" with the implemented system as its reference implementation, and the thesis/paper narrative: the framework was not designed on paper and then built — it was distilled from a working, measured stack.

6. State estimation without RL: the particle filter now runs in shadow mode

Per RQ's point (21 Jul, 10:59–11:00): transition probabilities are unknown, so optimising the MDP classically would demand RL — instead, the decision policy stays inside the QoE-analogy pipeline and the particle filter is used ONLY as the Manager's state estimator. Implemented today (`g1_particle_filter.py`, bridged behind `G1_M2_PF`, **shadow mode**: posteriors are computed and logged per sample as `meta2_pf`, no decision path is touched): 400 particles over a hybrid state — lid ∈ {open, closed}, **fill ∈ [0,1] (continuous — the reason this is a PF and not an exact HMM)**, sensing health, battery sufficiency, stuck. Observation models calibrated from the logged campaigns (spill rate vs motion excitation and freeboard; laser noise +65% while turning).

Replay validation over logged runs (no new experiments needed):

Run (logged)	Ground truth	P(lid open) final	Fill estimate

Real baseline 102149 (14 spills)	open, 246 g full	1.00	0.90 → 0.42 (14 spills ≈ 0.033 each — matches 7 g/spill / 200 g measured)
Real governed 171700 (12 marks)	open, 150 g	1.00	0.90 → 0.45 declining with marks
Real R2 abort 164422 (4 marks in 8 s)	open, fresh cup	1.00 by t≈8 s	front-loaded drop
Twin m2closedsim 125800 (0 spills, full motion)	CLOSED	0.00	0.91 (nothing spilled — correct)
Twin m2closedsim 130149 (0 spills)	CLOSED	0.00	0.90

RQ's "is the lid open or closed" is now an **inferred belief, not an operator setting** — discriminated correctly on all five replayed runs. Unified-pipeline hook (v2, after validating against the pending sealed-lid real runs R4): condition QoE expectations on the posterior — e.g. P(open) < 0.2 activates the covered-delivery analogy set — so the belief feeds the existing arbitration and the policy still never needs RL.

6a. Estimator per evidence density (RQ, 11:07–11:10) — settled by data, not argument

RQ's refinement: particle filters shine where samples are dense (navigation — laser returns, pose hypotheses), and are weak for sparse events (lid, battery); "no matter DST or particle filter or others, they are all probabilistic input, all feeding the decision model under POMDP." The stack now embodies exactly that: on the Worker side the K-of-N persistence filter + cell scores already are a sampled-evidence occupancy filter over dense returns; on the Manager side the 3-layer DST handles sparse events with explicit ignorance Θ. One empirical nuance worth adding to the discussion: for the lid, the ABSENCE of a spill under motion excitation is itself a Bernoulli observation at 2 Hz — dense evidence — which is why the shadow PF converged to P(open)=0.00 on the closed-lid twin runs from ZERO events. And because the PF runs in shadow while the DST layers run live, **both estimators now execute in parallel on every run over identical inputs**: Adrián's proposal ("we will try particle filter as the estimator, if it doesn't work… maybe the 3-layer DST is enough") becomes a measurable comparison on logged data rather than a design debate.